



Ginni Vishal 22203133

Fung Dao Wen 22230467

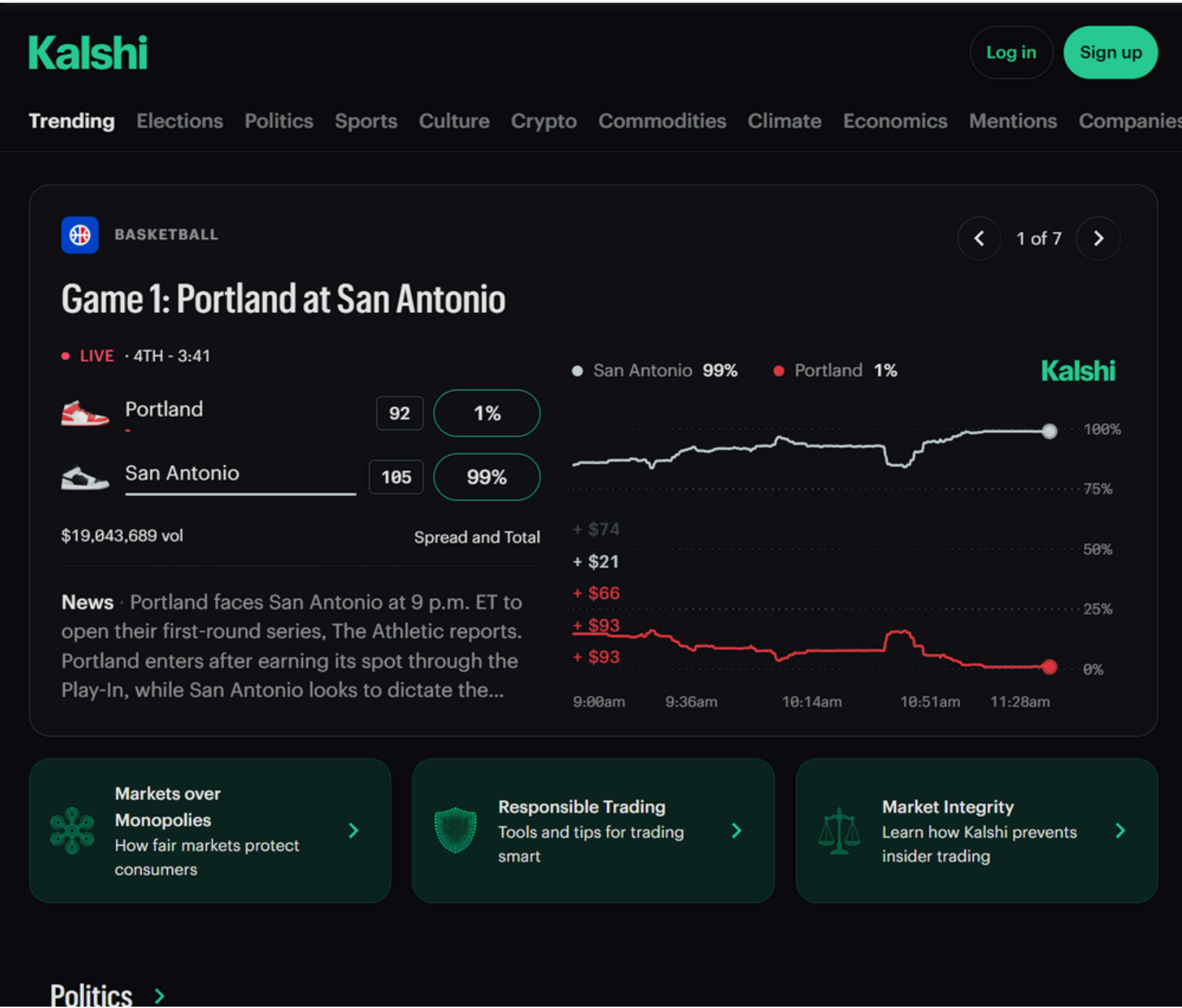
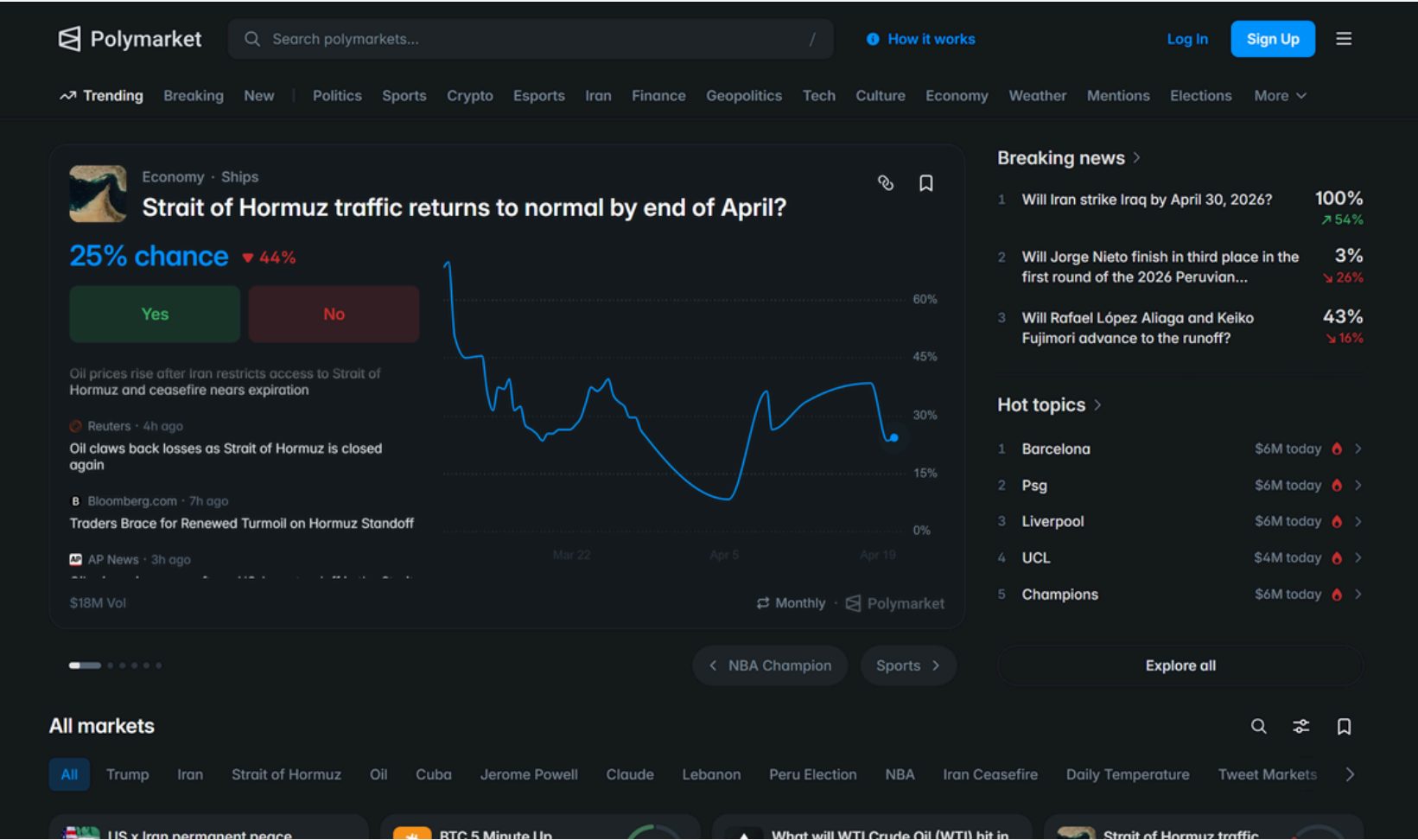
SADIEV Sherzodbek 22200959

Thit Htun LIN 23200073

THANT-THAZIN-EAIN 22202110

Let's Face It

Half of Polymarket is signal. Half is vibes. Traders can't always tell which is which, and doing the research manually is slow and scattered.



PROBLEM

SPECULATIVE

"Will X meme coin 5x?"

"Will celeb Y pump token Z?"

"Will vibes flip by Friday?"

✗ No public evidence base

✗ Hard to ground

✗ Weak audit trail

VS

RESEARCHABLE

"Will CPI print below X?"

"Will candidate A win state?"

"Will policy bill pass?"

✓ Evidence exists, but scattered

✓ News + filings + data sites

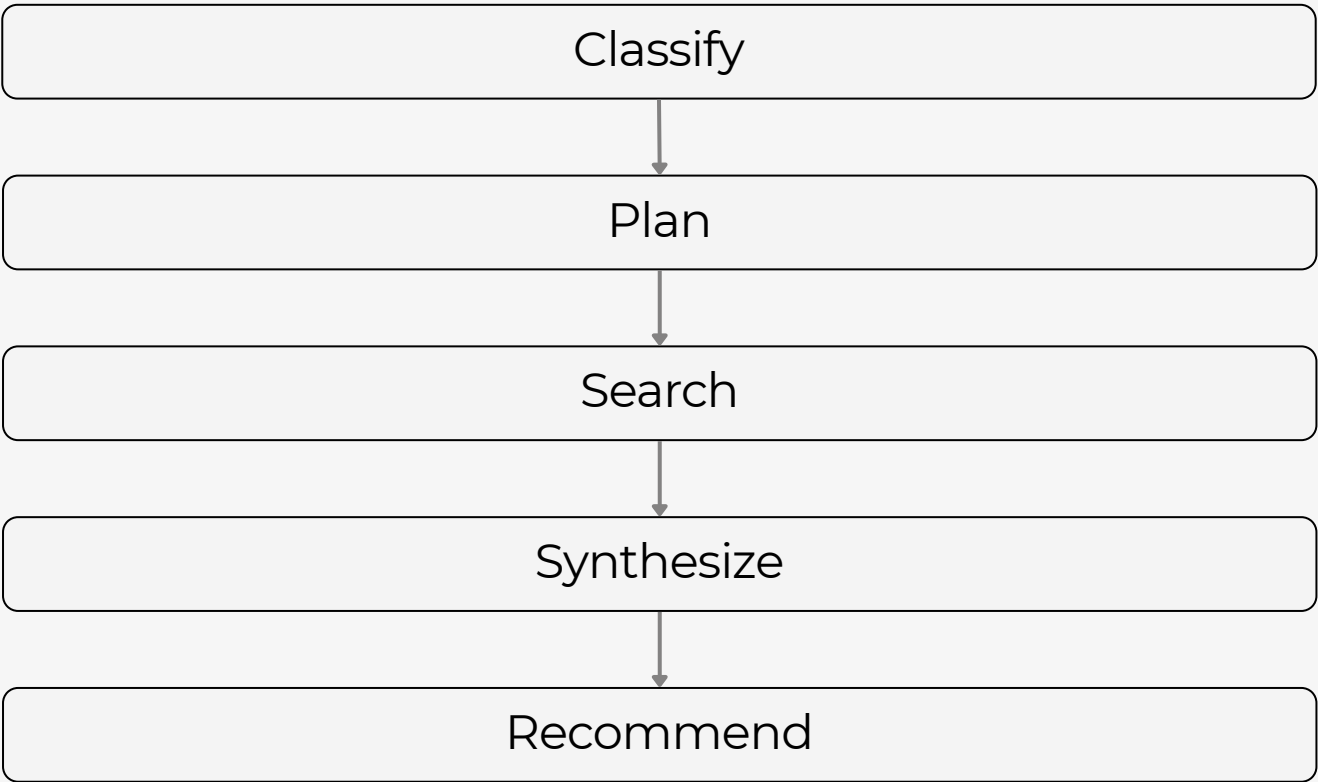
✓ Tabs/screenshots don't scale

Prediction markets are not the gap.
The research workflow is the gap.

Aperture = structured + auditable + durable pipeline
not a one-shot black-box answer

Introducing Aperture

A Convex-native research agent that decides whether a Polymarket event is researchable and, if it is, produces a sourced memo and a suggested market and side, with every step persisted and resumable.



Recommended markets

US x Iran permanent peace deal by June 30, 2026?

NO conviction 80

Why: No permanent peace deal exists, and the research highlights ongoing maximalist demands and failed negotiations. While the extended timeline provides more room for diplomacy, the lack of qualifying agreements or progress toward permanence remains a critical issue.

Risk: A prolonged negotiation process could eventually lead to a framework, but the deep mistrust and unresolved disputes make a permanent deal by this deadline unlikely.

US x Iran permanent peace deal by May 31, 2026?

NO conviction 85

Why: The absence of a permanent peace deal persists, with no qualifying agreements or official statements confirming one. The extended timeline allows for more negotiation time, but the fundamental lack of progress and maximalist demands remain significant barriers.

Risk: Prolonged negotiations could theoretically yield a framework, but the research indicates deep mistrust and unresolved disputes that make a permanent deal unlikely by this deadline.

US x Iran permanent peace deal by April 30, 2026?

NO conviction 90

Why: No permanent peace deal is in place, and the research indicates no progress toward one. The April 30 deadline provides slightly more time, but the lack of qualifying agreements or negotiations toward permanence remains a critical gap.

Risk: Extended negotiations or a surprise diplomatic breakthrough could emerge, though current trends suggest this is improbable given the failed talks and maximalist positions.

US x Iran permanent peace deal by April 22, 2026?

NO conviction 95

Why: No qualifying permanent peace deal exists as of April 19, 2026, per all research findings. The temporary ceasefire expires April 22, 2026, with no evidence of a permanent agreement being signed or confirmed by either government.

Risk: A last-minute breakthrough in the final hours before the deadline could invalidate the NO position, though current evidence suggests this is unlikely.

User Experience

How Aperture Works

Fast Scanning

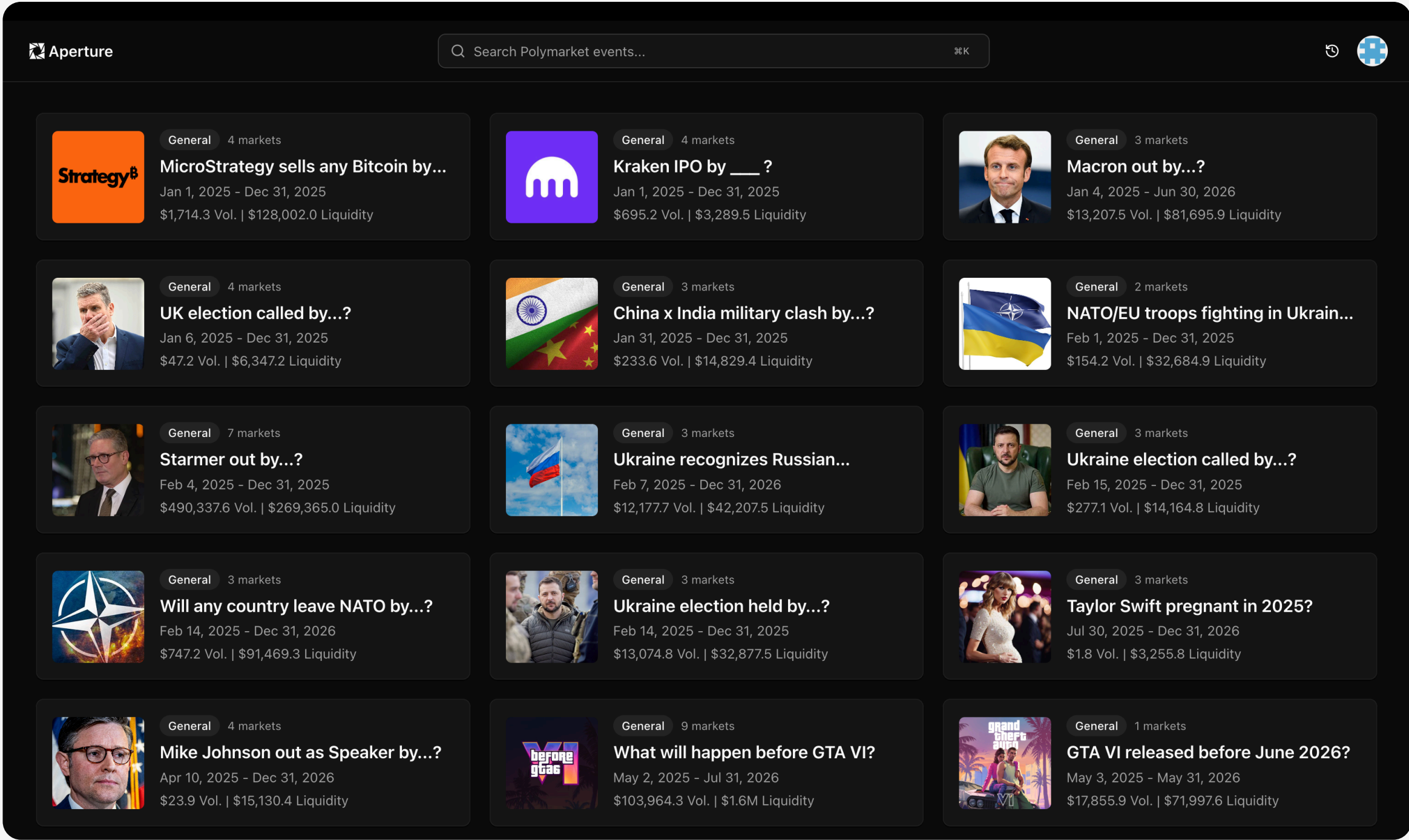
Users compare multiple markets at a glance.

Relevant Context

Tags, dates, volume, and liquidity stay visible.

Low Friction Flow

Discovery leads directly into the event page.



Scan

Read the dashboard quickly.

Compare

Check market context.

Enter

Open the selected event.

Entering the System

Research-first

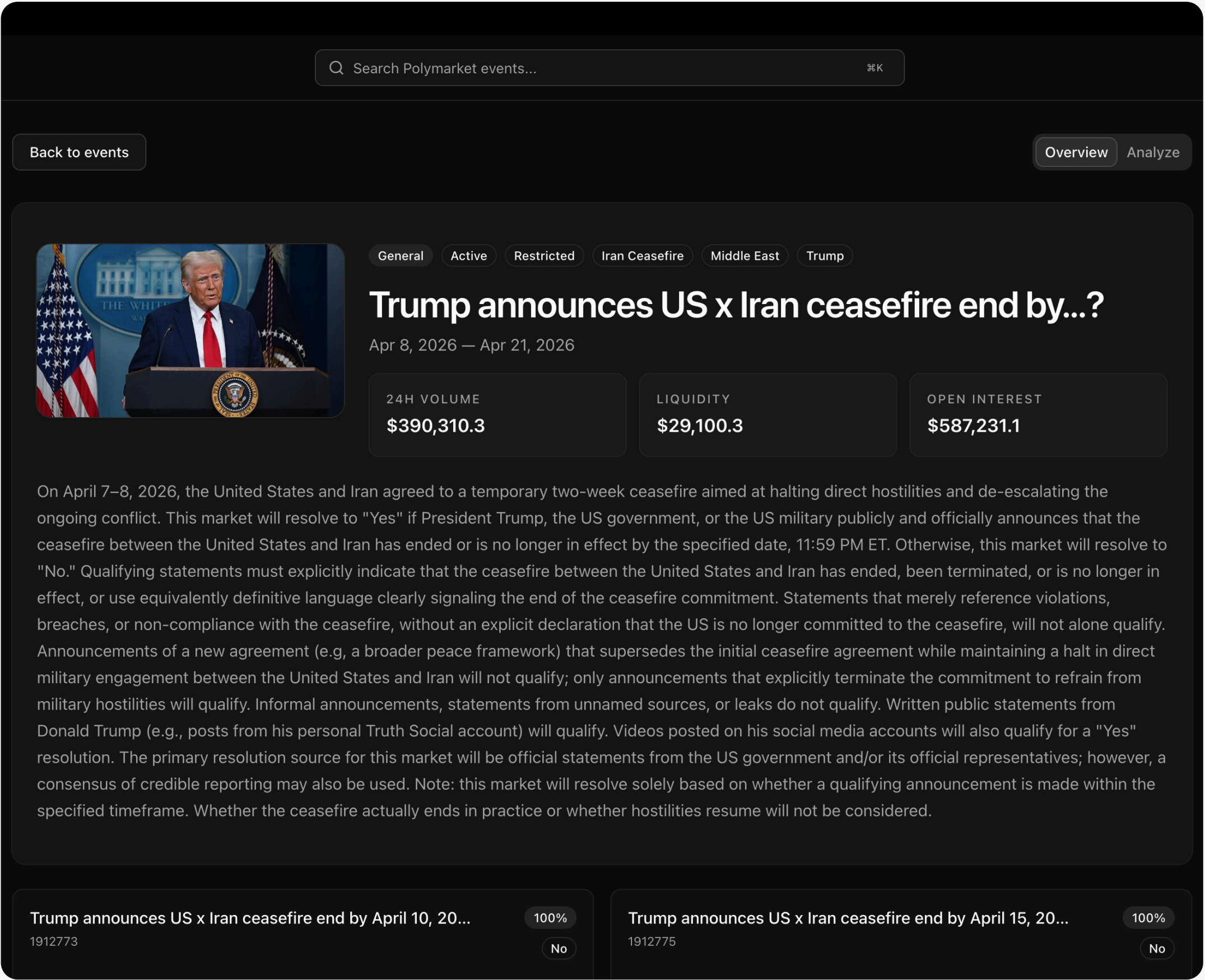
A structured interface for evaluating Polymarket events.

Event-centric

Each market opens into a dedicated event workspace.

Built for analysis

From discovery to evidence-based analysis.



Browse

Scan the market list.

Open

Review the event page.

Analyze

Move into deeper analysis.

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Analyze the Market

No WebSockets

No SSE route

Live DB UI

Click

Start Analysis

Convex Run

Update

Search Polymarket events...

Back to events

OverviewAnalyze



GeneralActiveFeaturedRestrictedIranTrumpceasefire

US x Iran permanent peace deal by...?

Apr 8, 2026 — May 31, 2026

24H VOLUME\$2.2M

LIQUIDITY\$1.2M

OPEN INTEREST\$5.4M

This market will resolve to "Yes" if Iran and the United states agree to a permanent peace deal by the specified date, 11:59 PM ET. Otherwise, this market will resolve to "No". A permanent peace deal refers to any agreement which explicitly indicates that military hostilities between the United States and Iran have ended or will permanently cease, or uses equivalent language clearly signaling a lasting end to military hostilities between the United States and Iran. Agreements that are explicitly temporary or which do not include a definitive agreement to end military hostilities between the US and Iran on a lasting basis (e.g. a temporary extension of the two-week ceasefire agreement announced on April 7, 2026), will not qualify. A qualifying agreement will be considered to have been established if either of the following conditions are met: - The United States and Iran each sign or formally adopt a written agreement (e.g. a treaty or multi-point agreement) which meets the above criteria. - Both the governments of the United States and Iran provide clear public confirmation that a qualifying agreement has been definitively established. Negotiations, statements of progress, or other statements which do not constitute a definitive announcement that a qualifying agreement has been reached will not count. The primary resolution source for this market will be official information from the governments of the United States and Iran; however, a consensus of credible reporting may also be used.

US x Iran permanent peace deal by April 22, 2026?

1919417

86%No

Volume	\$10.9M	Liquidity	\$592,152.6
Start date	Apr 8, 2026	End date	Apr 22, 2026
Last trade	14%	Best bid / ask	14% / 15%
Spread	1%	Status	Active

US x Iran permanent peace deal by April 30, 2026?

1919421

63%No

Volume	\$4.2M	Liquidity	\$356,032.5
Start date	Apr 8, 2026	End date	Apr 30, 2026
Last trade	37%	Best bid / ask	37% / 38%
Spread	1%	Status	Active

US x Iran permanent peace deal by May 31, 2026?

1919425

60%Yes

Volume	\$2.2M	Liquidity	\$144,753.0
Start date	Apr 8, 2026	End date	May 31, 2026
Last trade	59%	Best bid / ask	59% / 60%
Spread	1%	Status	Active

US x Iran permanent peace deal by June 30, 2026?

1962237

67%Yes

Volume	\$796,612.4	Liquidity	\$149,080.9
Start date	Apr 12, 2026	End date	May 31, 2026
Last trade	67%	Best bid / ask	65% / 68%
Spread	3%	Status	Active

Demo

aperture

Public

Unpin

Watch 0

Fork 0

Star 0

polymarket_agent

2 Branches

0 Tags

Go to file

T

Add file

<> Code

Spagestic	Update metadata in layout.tsx for improved project description and SEO	2ab8370 · 2 days ago	🕒 172 Commits
📁 .agents/skills	npx skills add vercel/ai		last week
📁 .claude/skills	npx skills add vercel/ai		last week
📁 .cursor/rules	drafted pdf link fetching pipeline		2 weeks ago
📁 app	Update metadata in layout.tsx for improved project descripti...		2 days ago
📁 components	history dropdown		2 days ago
📁 convex	history dropdown		2 days ago
📁 hooks	infinite scroll for dashboard		last week
📁 lib	refactor: expand Market and EventItem types with additional ...		4 days ago
📁 public	dashboard image added		2 days ago
📄 .env.example	api somewhat works		last month
📄 .gitignore	first commit		last month

About

Aperture is a Polymarket research analysis platform that classifies events, investigates open-web evidence, and produces structured market recommendations.

[aperturehk.vercel.app/](#)

- 📖 Readme
- 📄 MIT license
- 📈 Activity
- ★ 0 stars
- 👁 0 watching
- 🍴 0 forks

Releases

No releases published

[Create a new release](#)

Deployments

120

Libraries & Tools

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What Tech Stack Does Aperture Use?

FRONTEND

Next.js App Router

TypeScript

shadcn / Tailwind

streamdown

RUNTIME / DB

Convex

workflow

agent

reactive queries

AI / AGENT

AI SDK + Mistral

generateObject

Zod schemas

structured output

DATA / TOOLS

Firecrawl search

Firecrawl scrape

Convex actions

EXTERNAL INPUTS

Polymarket Gamma API

event + markets

page metadata

EXECUTION PATTERNS

Promise.all fan-out

per question parallelism

single structured helper

RESEARCH_BUDGET

governs iteration caps / scrape caps

bounded cost & latency

FRONTEND

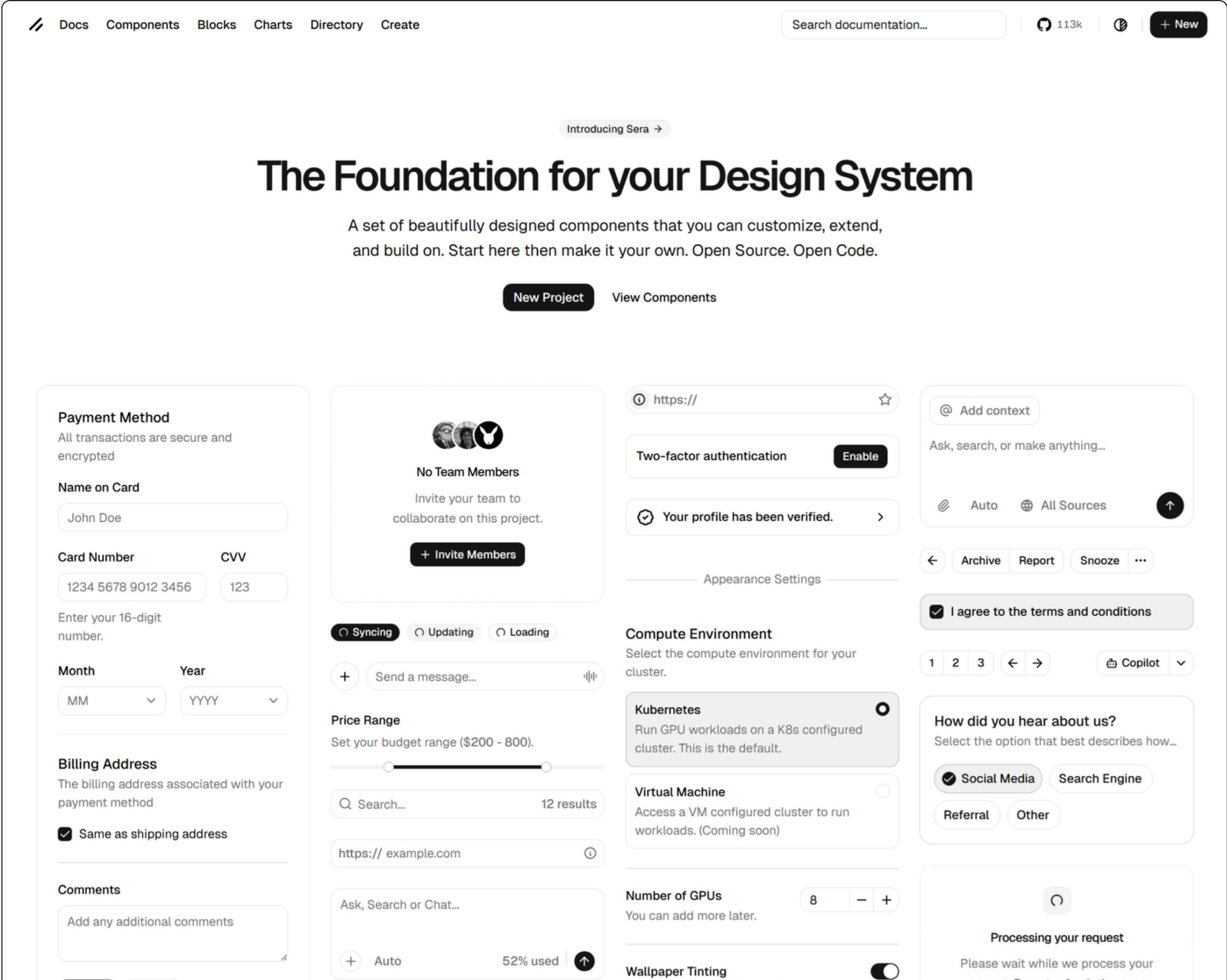
Next.js App Router

TypeScript

shadcn / Tailwind

streamdown

For streamed markdown, from general to specific



The backend platform that keeps your app in sync

[Start building](#)`> npm create convex`

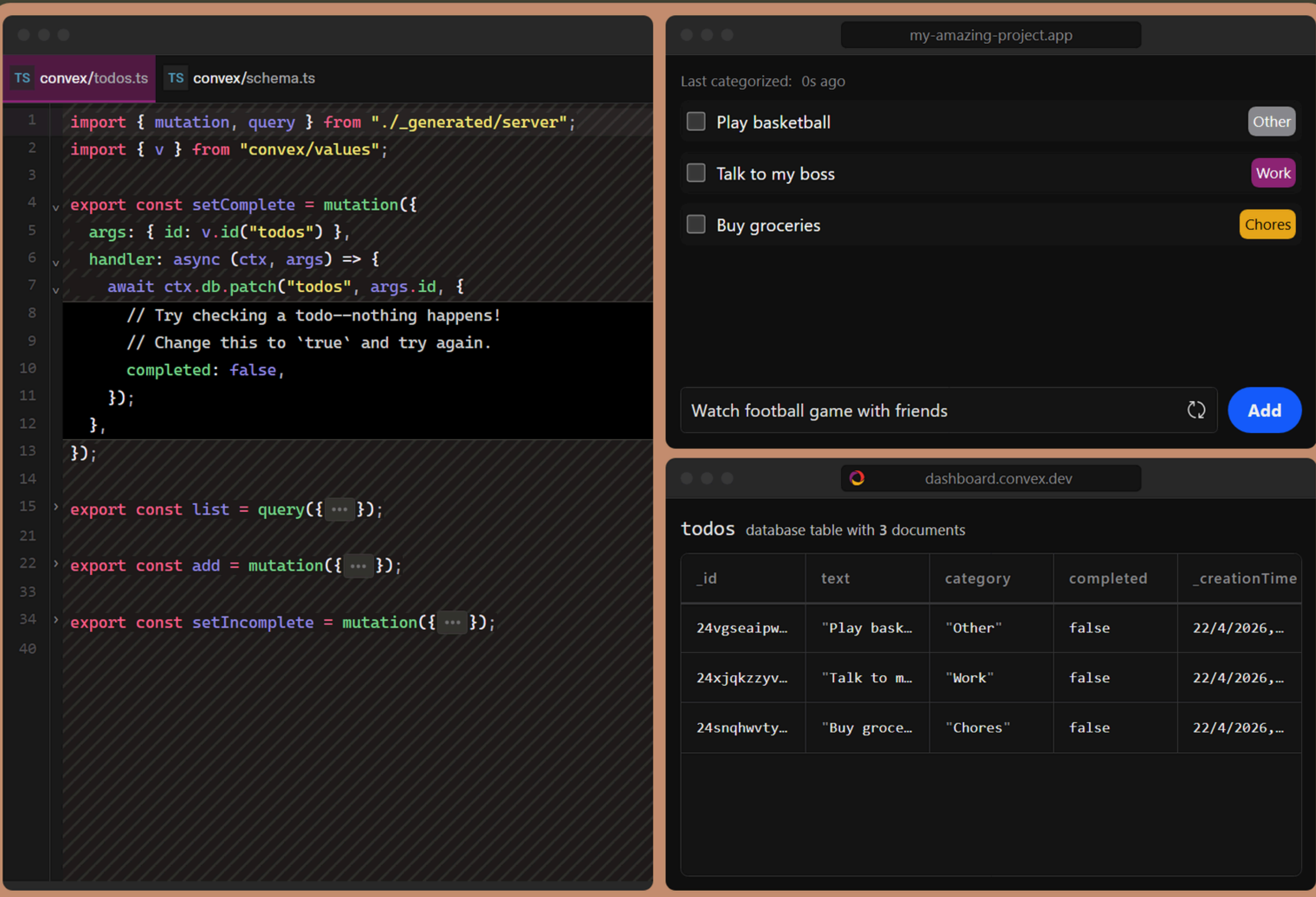
Everything is code

From database schemas to queries, from auth to APIs, express every part of your backend in pure TypeScript. Your backend code lives next to your app code, is typechecked and autocompleted, and is generated by AI with exceptional accuracy.

01

Always in sync

Backend built-ins



The screenshot displays the Convex development environment. On the left, a code editor shows TypeScript code for a todos application. The code defines a schema, a mutation to set a todo as complete, and a query to list todos. On the right, a UI mockup for 'my-amazing-project.app' shows a list of todos with checkboxes and category tags. Below the UI, a table view for the 'todos' database table shows three documents.

```
convex/todos.ts
import { mutation, query } from "../_generated/server";
import { v } from "convex/values";

export const setComplete = mutation({
  args: { id: v.id("todos") },
  handler: async (ctx, args) => {
    await ctx.db.patch("todos", args.id, {
      // Try checking a todo--nothing happens!
      // Change this to `true` and try again.
      completed: false,
    });
  },
});

export const list = query({ ... });

export const add = mutation({ ... });

export const setIncomplete = mutation({ ... });
```

my-amazing-project.app

Last categorized: 0s ago

- ☐ Play basketball Other
- ☐ Talk to my boss Work
- ☐ Buy groceries Chores

Watch football game with friends ↻ Add

dashboard.convex.dev

todos database table with 3 documents

_id	text	category	completed	_creationTime
24vgseaipw...	"Play bask...	"Other"	false	22/4/2026,...
24xjqkzzyv...	"Talk to m...	"Work"	false	22/4/2026,...
24snqhwvty...	"Buy groce...	"Chores"	false	22/4/2026,...

RUNTIME / DB

Convex
workflow ↔ For durable steps with retries

agent ↔ For persisted LLM threads

reactive queries

↕
For replacing any streaming endpoint

PRODUCT

Not just a database

Everything your product deserves to build, launch, and scale.

[Learn more](#)

SERVER FUNCTIONS

ACID DATABASE

VECTOR SEARCH

CRON JOBS

FILE STORAGE

REALTIME UPDATES

TYPE SAFETY

FRAMEWORK INTEGRATION

AUTOMATIC CACHING

STRONG CONSISTENCY


```
from pydantic import BaseModel
```

```
class Book(BaseModel):  
    name: str  
    authors: list[str]
```



```
import os  
from mistralai.client import Mistral  
  
api_key = os.environ["MISTRAL_API_KEY"]  
model = "mistral-8b-latest"  
  
client = Mistral(api_key=api_key)  
  
chat_response = client.chat.parse(  
    model=model,  
    messages=[  
        {  
            "role": "system",  
            "content": "Extract the books information."  
        },  
        {  
            "role": "user",  
            "content": "I recently read 'To Kill a Mockingbird' by Harper Lee."  
        },  
    ],  
    response_format=Book,  
    max_tokens=256,  
    temperature=0  
)
```



AI / AGENT

AI SDK + Mistral

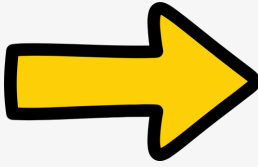
generateObject

Zod schemas

structured output

Zod-typed structured output

- Classify
- Judge
- Summarize
- Pick markets



```
# Accessed with `chat_response.choices[0].message.content`:  
print(chat_response.choices[0].message.content)  
# {  
#     "authors": ["Harper Lee"],  
#     "name": "To Kill a Mockingbird"  
# }
```

DATA / TOOLS

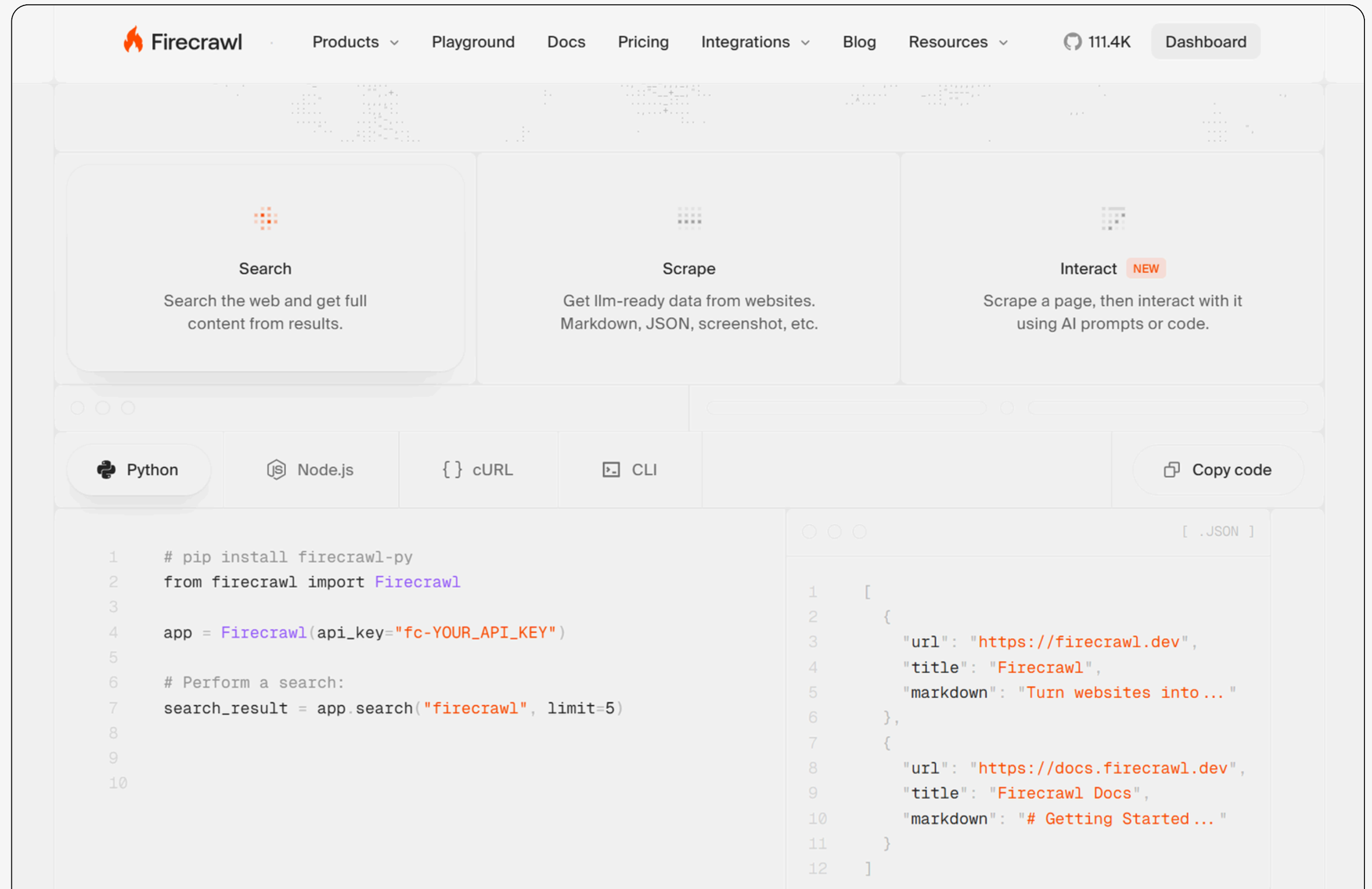
Firecrawl search

Firecrawl scrape

Convex actions

Leaf tools wrapped as Convex actions:

- *search* for discovery
- *scrape* for page content



Practical Uses of These Libraries & Tools

EXTERNAL INPUTS

Polymarket Gamma API
event + markets
page metadata

For *event + markets metadata* on the *page*

EXECUTION PATTERNS

Promise.all fan-out
per question parallelism
single structured helper

- For *fan-out* over questions and over scrape/summarize *per question*
- Every LLM call goes through a *single structured helper* with a Zod schema.

RESEARCH_BUDGET

Constant *governs iteration caps / scrape caps*

Bounded *cost & latency*

Limited API credits

COMPLEX LOGIC

How Does the System Handle Complex Flows?

pseudocode

```
if speculative: stop(reason)
qs = plan(event)
for q in parallel: research(q, budget=3)
answers = consolidate(qs)
picks = map_to_markets(answers, markets)
memo = synthesize(answers, picks)
```

(1) SPECULATIVE FILTER AS GATE

{ isSpeculative, reason }

A single decision short-circuits the whole workflow

(2) SUPERVISOR / SUB-AGENT LOOP

search → LLM judge (picks URLs to open (indices + reasons)) → scrape → summarize/relevance → refine

stop if relevantCount >= target OR iteration >= 3

Every transition is a row patch, so the UI shows progress without extra plumbing

(3) DURABILITY + AUDITABILITY

step.runAction per phase

retries on transient failure

full trail persisted in DB tables

Refreshing the page resumes the same run

(4) MARKET MAPPING

Map evidence → actual Polymarket market → produce a side + 0 - 100 conviction + rationale + key risk

The synthesizer doesn't just pick YES/NO

Extras & Limitations

(1) WHAT WE SHIPPED

auth-gated runs + per-user ownership + cancel button

Wired directly to workflow cancel.

debug log feed (researchLogs)

Transparent live view during the demo.

modular UI — analyze-panel/ split

Empty state, run header, memo, picks, question timeline, logs.

(2) WHAT WE DIDN'T

bounded by Firecrawl recall + LLM judgment

Paywalled or very recent sources can be missed.

research assistant — not trading advice

No execution; the user still makes the call.

budget caps can exhaust before evidence converges

Some questions end with insufficient public evidence.

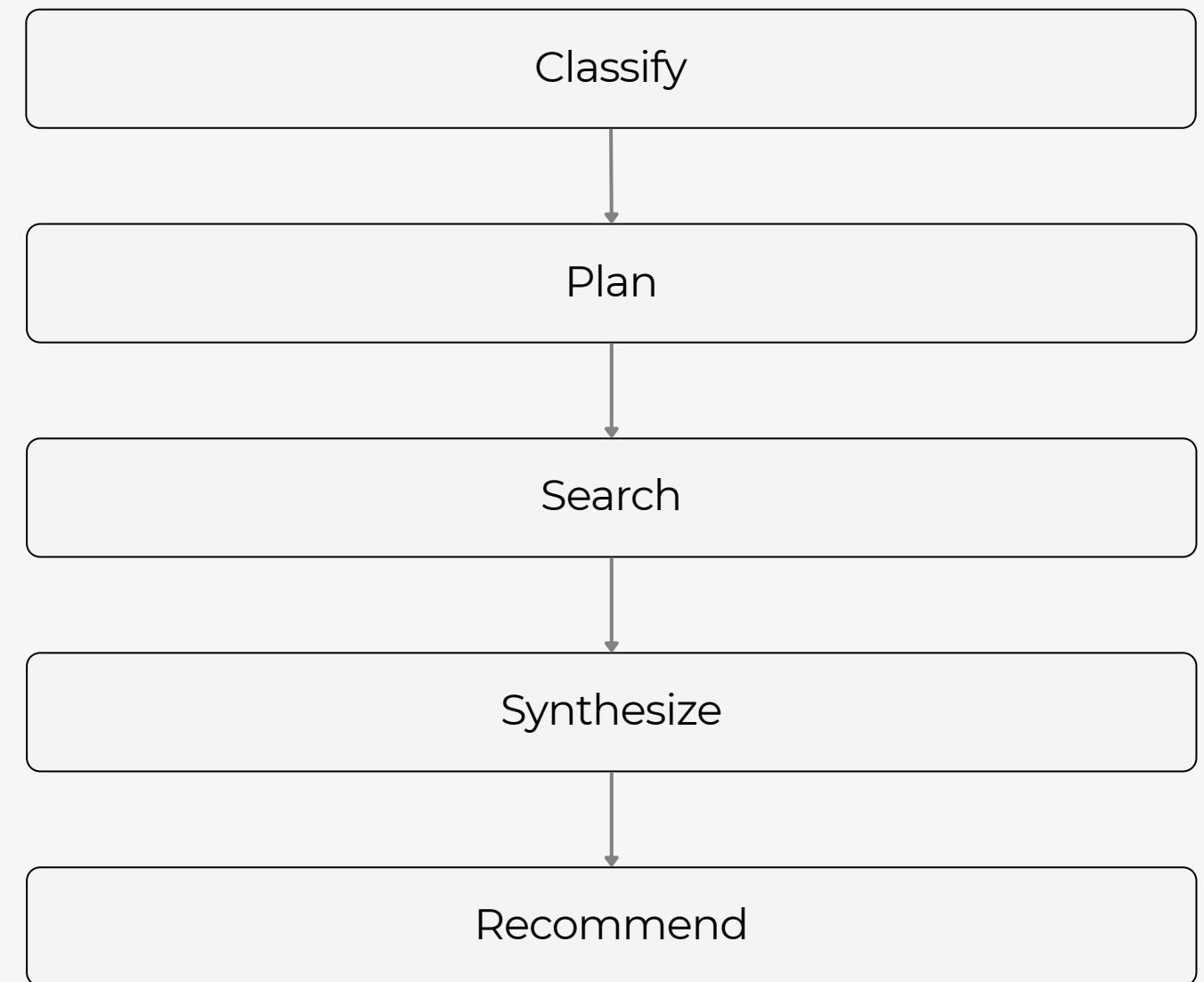
(3) WHAT'S NEXT

backtesting on resolved markets · recency-weighted scoring · citation-level UI · chat-with-this-memo via reused threads

Aperture

scattered web evidence → a durable, reactive research workflow.

We turned Polymarket event data and open-web sources into a pipeline that **classifies** the event, **cites** its sources, and **recommends** a market and a side.



the loop, end to end

thank you

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